DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CARE OF PATIENT WITH NEONATAL THROMBOCYTOPENIA
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

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1. PURPOSE

1.1. To provide information to health care providers on the clinical management neonatal thrombocytopenia.

1.2. To ensure that all medical and nursing staff are able to recognize early signs of thrombocytopenia for prompt management and prevention of complications.

2. DEFINITION & ABBREVIATION

2.1. **Thrombocytopenia** – is characterized by a low level of platelets in the blood below 150, 000/µL.

2.2. **Neonatal Allo-Immune Thrombocytopenia** – is a disease in which the mother produces antibodies against fetal platelet antigens inherited from the father.

2.3. **Neonatal Auto-Immune Thrombocytopenia** – is a disease in which the mother has idiopathic thrombocytopenia or systemic lupus in which maternal anti-platelet destroys fetal platelet.

2.4. **Infection-Mediated Thrombocytopenia** – is a type of thrombocytopenia wherein low platelet may cause from a reduced platelet production or accelerated platelet destruction due to infection.

2.5. **Mild Thrombocytopenia** – platelet count in the blood is between 100,000/µL to 150,000/µL.

2.6. **Moderate Severe Thrombocytopenia** – platelet count in the blood is between 50, 000/µL to 100,000/µL.

2.7. **Severe Thrombocytopenia** – platelet count in the blood is below 50,000/µL

2.8. **Platelets** – also called thrombocytes which are component of blood whose function is to stop bleeding by clumping and clotting blood vessels.

2.9. **Allo-antigens** – an antigen present only in some individuals capable of inducing the production of an antibody by individuals which lack it.

2.10. **PT (Prothrombin Time) and PTT (Partial Thromboplastin Time)** – is a blood test to help diagnose the cause of unexplained bleeding.

2.11. **Petechiae** – is fine red pinpoint rash over extremities which suggest bleeding.

2.12. **Ecchymosis** – is a subcutaneous spot of bleeding from extravasation of blood.

2.13. **IVIG** – Intravenous Immunoglobulin

2.14. **NEC** – Necrotizing Enterocolitis

2.15. **HIE** – Hypoxic-Ischemic Encephalopathy

2.16. **PPE** – Personal Protective Equipment

2.17. **NICU** – Neonatal Intensive Care Unit

3. RESPONSIBILITY

3.1. **Physicians (Consultant, Specialists and Residents)**

3.1.1. Responsible for assessing the patient and providing written orders in the medical file.

3.2. **Nurses (Head Nurse, Charge Nurse and Staff Nurse)**

3.2.1. Responsible to assess and monitor the patient with signs and symptoms of NAIT.

3.2.2. Carry out the procedures ordered to manage complications of neonatal thrombocytopenia.

3.2.3. Document all the procedures done in the medical file.

4. POLICY

4.1. All newborn patients presenting any signs of bleeding must be properly assessed and placed under close observation and will be subjected for admission to Neonatal Intensive Care Unit if indicated based on the newborn' clinical condition and as per physician's written order.

4.2. Assessment criteria for thrombocytopenia

4.2.1. Presence of petechiae shortly after birth

4.2.2. Ecchymosis

4.2.3. Hematuria

4.2.4. Blood in the stool

4.2.5. Mucosal bleeding

4.2.6. Thrombocytopenia

4.3. Classifications of Neonatal Thrombocytopenia

4.3.1. Immune-Mediated Thrombocytopenia

4.3.1.1. Neonatal All-Immune thrombocytopenia

4.3.1.2. Neonatal Auto-Immune Thrombocytopenia

4.3.2. Infection-Mediated Thrombocytopenia

4.3.2.1. Bacterial Infections

4.3.2.2. Viral Infections

4.3.2.3. Fungal Infections

4.3.3. Miscellaneous Causes of Thrombocytopenia

4.3.3.1. Drug-related

4.3.3.2. Disease-related (e.g. NEC, Birth Asphyxia)

4.4. Diagnosis

4.4.1. It is based on the clinical and serological findings, including a low platelet count and alloimmune antibodies. Blood tests such as complete blood count, PT, PTT, Peripheral Blood Smears, Liver enzymes, CRP and blood cultures are indicated.

4.4.2. Serial platelet counts should be obtained for the first 5-7 days after delivery to keep the

platelet counts higher than 30,000/µL without active bleeding and higher than 50,000/µL – 100,000/µL with active bleeding.

4.5. Medical Management

4.5.1. Platelet Transfusion

4.5.1.1. If the neonate presents with clinical bleeding or the platelet count is less than 30,000/ µL, then platelet transfusion is indicated.

4.5.1.2. Following the transfusion of platelet an incremental platelet count should be measured approximately one hour after the transfusion is complete and subsequently at least daily until the platelet count has been demonstrated not to be falling.

4.5.2. Intravenous Immunoglobulin Transfusion

4.5.2.1. Intravenous Immunoglobulin may be considered but should not be used as an alternative to platelet transfusion, since the action of IVIG is observed after 18 hours and platelet counts don't rise before 24-48 hours.

4.5.2.2. Neonates with moderately severe thrombocytopenia (30,000/µL – 50,000/µL) without obvious bleeding can be treated with IVIG alone with a total dose of 2g/kg which are given over 2-5 days.

4.5.2.3. In platelet-transfused patients, IVIG at 0.14-1 g/kg/day for 2-5 days can be given to potentially prolong the survival of the incompatible platelets.

4.5.3. Cranial Ultrasonography examination is indicated for every newborn with significant thrombocytopenia to exclude intracranial hemorrhage.

5. PROCEDURE

5.1. The physician will assess the patient with signs of thrombocytopenia.

5.2. The physician will write the order of the procedures in the medical file according to his management.

5.3. The nurse will verify and carry out physician's written order.

5.4. Perform hand hygiene and wear appropriate PPE.

5.5. Identify the patient correctly.

5.6. The nurse will perform the procedures to manage thrombocytopenia (e.g. blood tests, IVIG infusion, and cranial ultrasound) according to physician's order and management.

5.7. The nurse will provide the necessary nursing care indicated according to physician's management.

5.8. Perform continuous re-assessment of the patient and refer any abnormal observations to physician immediately.

5.9. If the patient requires blood transfusion, transfuse blood according to policy and written physician's order.

5.10. Re-evaluated and monitor the patient's vital signs closely.

5.11. Document all the procedures done in the medical file.

6. MEASUREMENT

6.1. 100% of the staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0019(v3): Care of Patient with Neonatal Thrombocytopenia, 28/05/2023, p. 1-5

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10 p.126-127

7.3. https://www.medscape.org/viewarticle

7.4. https://www.ncbi.nlm.nih.gov.pmc/articles

8. ATTACHMENT

8.1. None

9. REVISION CLAMSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.